Harshini Balamurugan Vadivazhagi

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Education

Master of Science in Computer Science Aug 2025 – May 2027
Arizona State University, Tempe, AZ ***GPA: 3.94***

Bachelor of Technology in Computer Science Sep 2021 – May 2025
Vellore Institute of Technology, Chennai, India ***GPA: 3.93***

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, Kotlin, Java (coursework), C++ (coursework)

Backend & Data: Node.js, Fastify, Express.js, Prisma, PostgreSQL, PostGIS, Docker, Microservices, REST APIs, Event-Driven Architecture (pg-boss), JWT/OAuth

AI & ML: LLM agents, LangChain, RAG, prompt engineering, evaluation harnesses, PyTorch, semantic segmentation (U-Net), Faster R-CNN, OpenCV, scikit-learn, time-series forecasting

Frontend: React/Next.js, Streamlit, Bootstrap, Kotlin/Jetpack Compose

Practices: Git, unit/integration/end-to-end testing, Vitest, Linux, code review practices

Professional Experience

Software Engineering Intern June 2026 – Aug 2026
Optiwise – AgriChoice (Farm-Equipment Rental Marketplace) *Remote*

- Built a rental-extension billing feature end to end across four TypeScript/Fastify production services and a Kotlin/Jetpack Compose Android client; delivered GST-compliant document generation in four languages and regenerated 684 stored records at scale.
- Engineered a four-service microservices backend (identity/auth, catalog & bookings, payments/escrow, background worker) reaching approximately 2,400 passing automated tests, using async event-driven communication (pg-boss) in TypeScript, Node.js, Fastify, Prisma, and PostgreSQL/PostGIS.
- Built a cursor-paginated admin REST API with bulk account moderation and audit logging, and eliminated double-booking risk across a 13-state booking state machine using optimistic concurrency control, partial-unique-index constraints, and shared Zod DTO contracts.
- Diagnosed three stacked production defects and five client-server API contract mismatches missed by the existing test suite; completed a 20-scenario manual test pass across four user roles and drove 42 commits of fixes through rigorous code review.
- Deployed the platform to a Fly.io staging environment across all four services and validated the native Android client against the backend by exercising the farmer booking journey through an offline-first booking queue.

Machine Learning Research Intern Aug 2023 – Sep 2023
Interlace India Pvt. Ltd. *On-site*

- Built a real-time email-classification pipeline in Python; mitigated class imbalance through keyword weighting and threshold tuning to reduce false negatives, and communicated findings to stakeholders.

Other Work Experience

Graduate Service Assistant – Grader May 2026 – Present
Arizona State University, School of ECEE *Tempe, AZ*

- **Grading**: Evaluated and analyzed student coursework for four sections of the Random Signals curriculum, applying structured rubrics and delivering targeted technical feedback to address conceptual gaps

Projects

Where Did the Forest Go? Amazon Deforestation Mapping *Python, PyTorch, U-Net, Sentinel-2*

- Built a Sentinel-2 + U-Net (PyTorch) pipeline mapping Amazon deforestation in Rondonia, Brazil, estimating 6,800 ha of forest cleared from 2021 to 2026 (95% CI 6,200 to 7,500 ha) through blind, stratified validation on 600 hand-labelled points.
- Diagnosed false alarms as a training-label problem: a SWIR-band ablation gave no gain, while retraining on MapBiomas labels instead of ESA WorldCover raised precision for cleared forest from 59% to 88% (92% overall accuracy).
- Trained a U-Net from scratch on a Kaggle GPU with a spatially held-out split to prevent leakage, reaching 0.91 test IoU versus 0.85 for an NDVI-threshold baseline.

Controlled Persistent Memory for LLM Assistants *Python, RAG, Benchmarking, Retrieval Fusion*

- Co-authored a research paper designing a controlled-memory architecture with a write router, versioning, and multi-signal retrieval to improve reliability in AI-agent systems without clear precedent.

- Implemented deduplication and versioning logic and fused four retrieval signals into one pipeline, independently navigating an unfamiliar research problem from first principles.

DispatchIQ: AI Fleet Command Center *Next.js, TypeScript, Groq LLM, Google Maps, Twilio*

- Built an AI fleet dispatch platform integrating Groq LLM (llama-3.3-70b) with the Trucker Path NavPro API to auto-score and recommend drivers by Hours-of-Service, GPS proximity, cost/mile, and safety, with full reasoning shown per recommendation.
- Engineered a custom dynamic-programming knapsack algorithm for load/route optimization, selecting the most profitable combination of loads within weight and Hours-of-Service constraints.
- Cut billing processing time from 45 minutes to 90 seconds by building a Groq Vision (llama-3.2-11b) OCR pipeline that extracts BOL/POD/receipt data and auto-generates invoices.

Clickless: Agentic LLM-Driven Buying Guide *Python, LangChain, OpenAI API, LLM Agents*

- Designed and built a three-agent Planner–Retriever/Ranker–Explainer pipeline in Python, processing 25–30 GB of product metadata into a query-efficient index and achieving 92% precision@50.
- Combined deterministic rule-based logic with LLM reasoning to reduce hallucination risk, applying professional judgement to validate AI-generated outputs across multi-turn product refinements.

Displaced Voices: Data Science for Housing Justice *Time-Series Forecasting, Regression, Python*

- Contributed to a five-person predictive-modeling track forecasting Arizona eviction filings via time-series modeling, projecting 54,143 evictions for 2025 – a 16% increase over 2022 – to support proactive state rental-assistance budgeting.
- Analyzed three years (2022–2024) of Maricopa County eviction records to identify seasonal risk windows, flagging summer months as consistently highest-risk for early intervention.

Livestock Protection: AI Cattle Detection & Predator Alerts *Faster R-CNN, Deep CNN, OpenCV*

- Designed a real-time livestock monitoring pipeline using Faster R-CNN and generative-AI-based preprocessing to automatically count cattle and detect predator intrusions from farmland camera imagery.
- Built a five-module system (image capture, GenAI preprocessing, cattle counting, intrusion detection, real-time alerts) that classifies cattle versus predators and triggers instant SMS/app alerts to farm owners.

LinkedIn Post Generator *LangChain, Groq LLM, Few-Shot Prompting*

- Built an AI-powered Streamlit app generating on-brand LinkedIn posts from a topic, using few-shot prompting over a curated dataset of real posts via LangChain and the Groq LLM.
- Implemented a tag-based few-shot retrieval system that pulls matching example posts from a preprocessed dataset to improve output realism and consistency.

MeloLearn: Tuning Careers Through Music *HTML, CSS, JavaScript, Bootstrap, Express.js*

- Developed a full-stack music education and creation platform with a content-ranking system evaluating originality, creativity, and technical proficiency, surfacing a live "Top 10" leaderboard to drive engagement.
- Implemented audio editing tools (trim, reverse, speed) and a Slick.js carousel UI to streamline music browsing and remix workflows across curated course content.

PUBLICATIONS

OTP Verification & Zero Trust Security for RBAC in Medical Records ***ICSTSN 2024***

- Proposed a Zero Trust architecture combining OTP verification, RBAC, salted password hashing with bcrypt/Argon2, JWT session tokens, and AES/SHA-256-encrypted audit logs for HIPAA-aligned, tamper-evident hospital-record access; published at ICSTSN 2024 and cited in subsequent Zero Trust Architecture research.